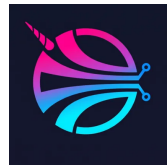


Anyflo DeFi Provider

Anyflo in one slide



Non-custodial stablecoin payment orchestration for businesses.

- Customers create payment intents; independent Solvers bid to execute them.
- Existing Gateway, Coordinator and Solver Hub coordinate access and order state.
- Chain receipts establish funding and delivery; Anyflo does not custody the funds.

Business use case: paying a supplier

Scenario: A business holds bridged USDT on Base; its supplier wants native USDC on Base.

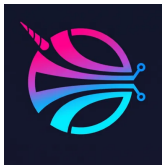
- The business wants a clear quote and delivery without managing a swap.
- The supplier wants the agreed asset at the specified address.
- The payment provider needs policy checks and evidence for its obligations.

User story: quote to supplier delivery

As a business customer, I want to pay a supplier in USDC using my USDT, so the supplier receives the agreed amount without my team operating DeFi.

1. Submit the payment intent and receive a solver bid.
2. Fund the accepted order; the Solver verifies the deposit.
3. The Solver swaps through its selected source and delivers USDC.

The Solver: from inventory to liquidity access



A Solver is the **liquidity and execution agent**, not a new payment network.

The existing Solver needs payout inventory. This project adds a standalone agent that uses verified customer funding to obtain the payout asset through **Uniswap AMMs or a custom Aqua position**.

One configured EVM pair, fixed input, one strategy and one active order.

What we built for this hackathon

- Public standalone Solver: existing JWT, signed bids/acceptance and CBOR API.
- Uniswap API quotes, bounded Permit2 authorization and validated AMM swaps.
- Permissionless Aqua/SwapVM position app, maker lifecycle tools and execution.
- Durable spend checkpoints and a runnable mock-Hub, real-contract demo.

Why compliance must cover the route

Payment providers may need to meet sanctions, AML and internal risk obligations. Screening the sender and recipient alone does not assess how funds move.

- An AMM route can involve multiple tokens, pools and venues.
- A payment provider needs an auditable route decision consistent with policy.
- Requirements depend on jurisdiction, customer and applicable law.

Today: Anyflo screens counterparties; Solver route checks are future work.

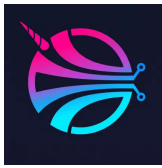
Why check an AMM route before bidding?

The Solver can inspect a quoted route **before committing** to an order:

1. Request an AMM quote with expected output and protected minimum.
2. **Future:** evaluate the proposed assets, pools and venues against policy.
3. Bid only if the route is acceptable and the economics meet the order.

Future: recheck the refreshed route before execution; quotes can change.

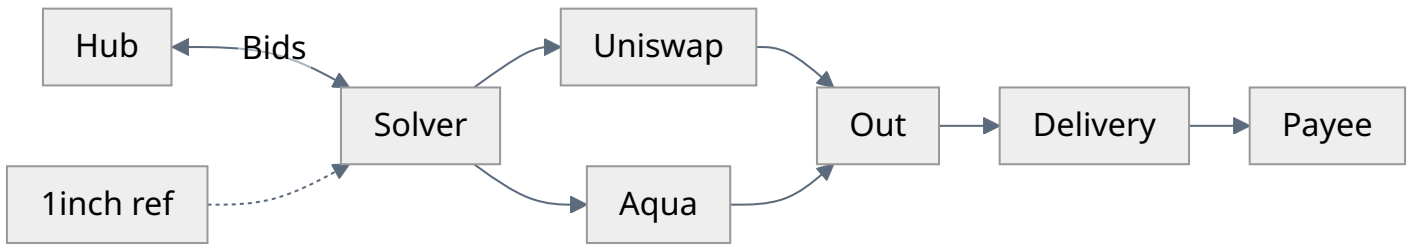
How on-chain liquidity funds the payout



- **Uniswap:** liquidity providers deposit tokens into AMM pools; the Solver swaps against a quoted pool route.
- **Aqua:** a maker configures a wallet-backed position; SwapVM quotes and executes against that position.
- After deposit verification, the Solver swaps its net input and uses the output for the agreed DeliveryAccount payout.

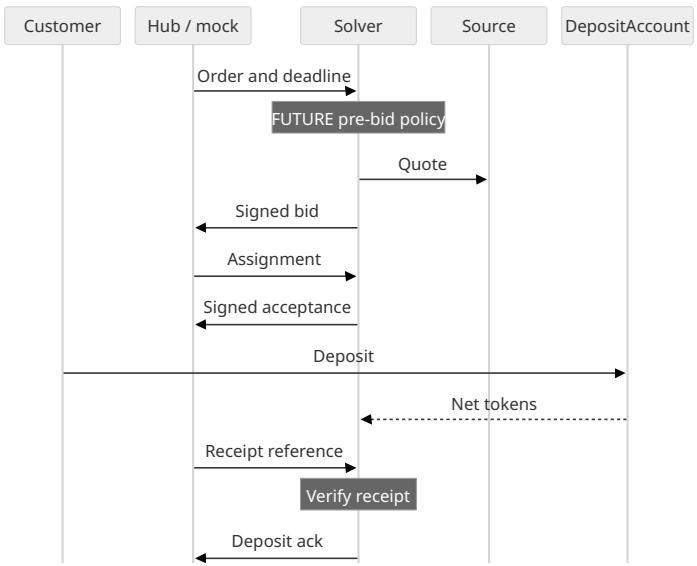
Pool reserves and maker backing can change; neither a quote nor an API response guarantees liquidity.

Integration: choose a source, preserve Anyflo settlement



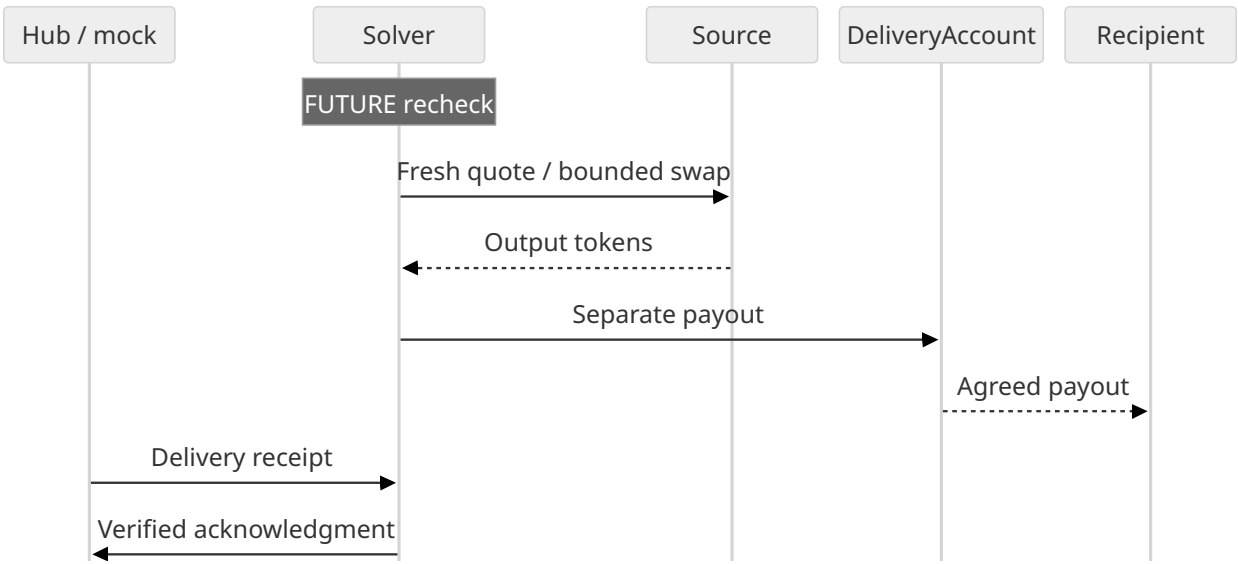
Aqua’s executable quote is on-chain. The 1inch HTTP quote is **reference data**. No resolver enrollment, global position crawler or provider fallback.

Swimlane: auction and funding



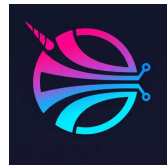
The Hub is mocked. Chain evidence is real. **FUTURE** policy is not implemented.

Swimlane: execution and delivery



Accepted payout stays fixed. Swap and delivery are separate transactions.

Aqua app: a wallet-backed programmable position



Position: **0.99–1.01 band**, **5 bps input fee**, **30-second decay**.

- A separate maker ships a permissionless position on the official SwapVM engine.
- Virtual reserves refer to actual maker-wallet balances and registry allowances.
- Real ship, fill and dock receipts were observed; docking leaves custody intact.

Observed local payment results

Strategy	Input	Swapped	Delivered USDC	Checks
Uniswap V3	10.000000	9.990000	9.934087	11
Aqua / SwapVM	10.000000	9.990000	9.934580	14

Both start with **zero Solver token balances**. Separate protocol fee: 0.100000 test USDT; retained input premium: 0.010000. These are one local run, not rates.

Added value

Anyflo and business customers: more liquidity paths without stocking every payout asset in each Solver wallet; the existing order API stays unchanged.

Uniswap and Aqua: business-originated payment flow and a potential use case connecting DeFi liquidity to regulated payment rails.

LP capital, execution risk and applicable legal obligations remain.

Compliance roadmap — explicitly not built here

Existing production context: Anyflo access controls and counterparty screening. The mock demo does not execute those backend checks.

Future Solver goal: inspect the complete execution plan, every route, asset, venue and counterparty **before bidding and again before executing**. A changed quote or route must be checked again.

Transaction-bound validation is implemented; **route AML/policy enforcement is not**.

Evidence, boundaries and next steps

Verified: 43 Bun tests, 5 native Forge settlement tests, both real local protocol flows, and live read-only API probes. Source pins were also verified.

Mocked: Hub/customer role, local provider HTTP and Anyflo settlement doubles. **Not verified:** production backend integration or mainnet settlement.

Next: production Solver modularization and separately scoped route compliance. UniswapX remains future research only.